

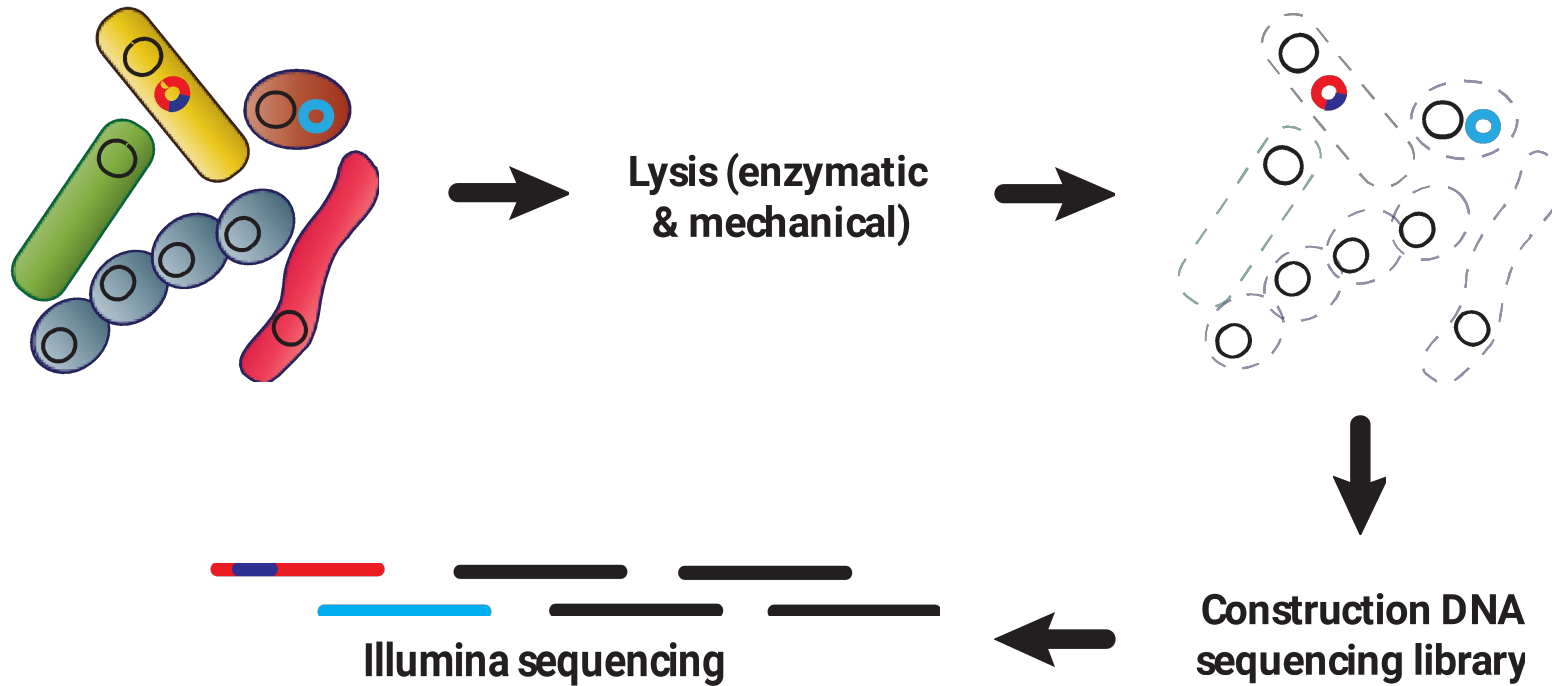
Reference alignment-based methods in metagenomics

Puspendu Sardar

University of Cambridge

2025.11.17

Basic laboratory workflow



Reference based alignment

WE_HAVE_LONG_SEQUENCE_TO_ALIGN_OUR_READS_TO

ECE ADS_TO ONG_S IGN_ G_SEQ
OU EADS ALOG WE TO_A
E_LON UR_R HARVE ENCE
AVE_L EQU _READ LON
LIGN _HAV _ALIG _TO
E_TO

Reference based alignment

WE_HAVE_LONG_SEQUENCE_TO_ALIGN_OUR_READS_TO

WE_ LON ECE LIGN ADS_TO
 _HAV ONG_S E_TO IGN_ _READ
 AVE_L G_SEQ TO_A _OU EADS
 E_LON EQU _ALIG UR_R

Reference based alignment / problems 1

WE_HAVE_LONG_SEQUENCE_TO_ALIGN_OUR_READS_TO

WE_ HAV_ ONG_S E_TO IGN_ _READ
AVE_L G_SEQ TO_A _OU EADS
E_LON EQU_ _ALIG UR_R

- What do we do with the reads: “ALOG”, “HARVE”, “_TO” and “ENCE”?
- Why do we have a few (or a lot of) reads that are very different?
- How to define rules to make this automatic and quick?
- Can you use the same rules every time?

Reference based alignment / problems 2

- You are using a too distant reference sequence from the target
- You have another too close reference sequence
- You have an incomplete / fragmented reference sequence
- You have a closely related (highly homologue) species in the sample
- Mobile genetic elements

Coverage

WE_HAVE_LONG_SEQUECE_TO_ALIGN_OUR_READS_TO

WE_HAV_ONG_S_E_TO_IGN_READ_TO
AVE_L_G_SEQ_TO_A_OU_EADS
E_LON_EQUE_ALIG_UR_R

112122223332222212121222223322121222332111

$\text{sum}(\text{nucleotide level read coverage}) / \text{length}(\text{reference})$

-> mean coverage (1.9)

Ideal world 1 – reference-based re-alignment

- We know all bacterial genomes that are potentially present in the analysed community and have good references for each of them
- Align all short reads to reference sequences (bowtie2, bwa)
- Extract unique alignment information (presence / absence)
- Extract coverage information (abundance)
- Extract SNP and short indel information (genetic variance)
- Extract structural variations (genomic rearrangements)
- Gold standard whole-genome re-alignment pipelines

Ideal world 1 – reference based re-alignment (potential real world examples)

- You are testing different probiotics in the relation of the presence / abundance of an unwanted bacterial species in the human colon (colonisation resistance)
- Database build-up: the unwanted bug, all probiotic species, a few well known commensals from human colon microbiota
- Take multiple samples from patients during the probiotic treatment and check the good (commensal & probiotic) / bad (unwanted) bacteria ratio
- Extreme but clinically relevant example: FMT (faecal microbiota transplant) in *Clostridium difficile* infection

De novo assembly vs. realignment

	De novo assembly	Reference based realignment
Data source	Short read next generation sequencing data	
Preprocessing	Preprocessing is very similar (QC, duplication removal, merge overlapping)	
Processing	Slow and resource hungry	Quick and lower resource need
Post-processing	Looking at the features of the genome	Looking at the differences to the reference
End product	Sequence file, gene annotation	Alignment file, variant calling
Sensitivity	+	+++
Quantification	++	+++
Novel findings	++++	+
Price	++++	++
Knowledge need	+++	+++

Complementing each other

- Finding genes/sequences of interest -> realignment
- Extending gene sequences with flanking regions -> de novo approach
- Filter out unwanted short reads -> realignment
- Do the de novo genome assembly
- Combine them iteratively to refine your final de novo assemblies

Short read aligner software

- $\textit{FastaFile}_{genome} + \textit{SequenceReads}_{sample} = \textit{Alignment}$

Short read aligner software

- $\text{FastaFile}_{\text{genome}} + \text{SequenceReads}_{\text{sample}} = \text{Alignment}$
- $(\text{FastaFile}_{\text{genome}} + \text{SequenceReads}_{\text{sample}}) \text{Options} = \text{Alignment}$

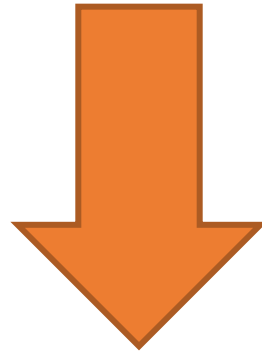
Short read aligner software

- $\text{FastaFile}_{\text{genome}} + \text{SequenceReads}_{\text{sample}} = \text{Alignment}$
- $(\text{FastaFile}_{\text{genome}} + \text{SequenceReads}_{\text{sample}})\text{Options} = \text{Alignment}$
- $\frac{(\text{FastaFile}_{\text{genome}} + \text{SequenceReads}_{\text{sample}})\text{Options}}{\text{NumberOf}_{\text{CPU}}} = \text{Alignment}$

Short read aligner software

- $\text{FastaFile}_{\text{genome}} + \text{SequenceReads}_{\text{sample}} = \text{SAMFile}$
- $(\text{FastaFile}_{\text{genome}} + \text{SequenceReads}_{\text{sample}})\text{Options} = \text{SAMFile}$
- $\frac{(\text{FastaFile}_{\text{genome}} + \text{SequenceReads}_{\text{sample}})\text{Options}}{\text{NumberOfCPU}} = \text{SAMFile}$
- $\frac{(\text{IndexedFasta}_{\text{genome}} + \text{SequenceReads}_{\text{sample}})\text{Options}}{\text{NumberOfCPU}} = \text{SAMFile}$

Short read aligner software



Short read aligner software

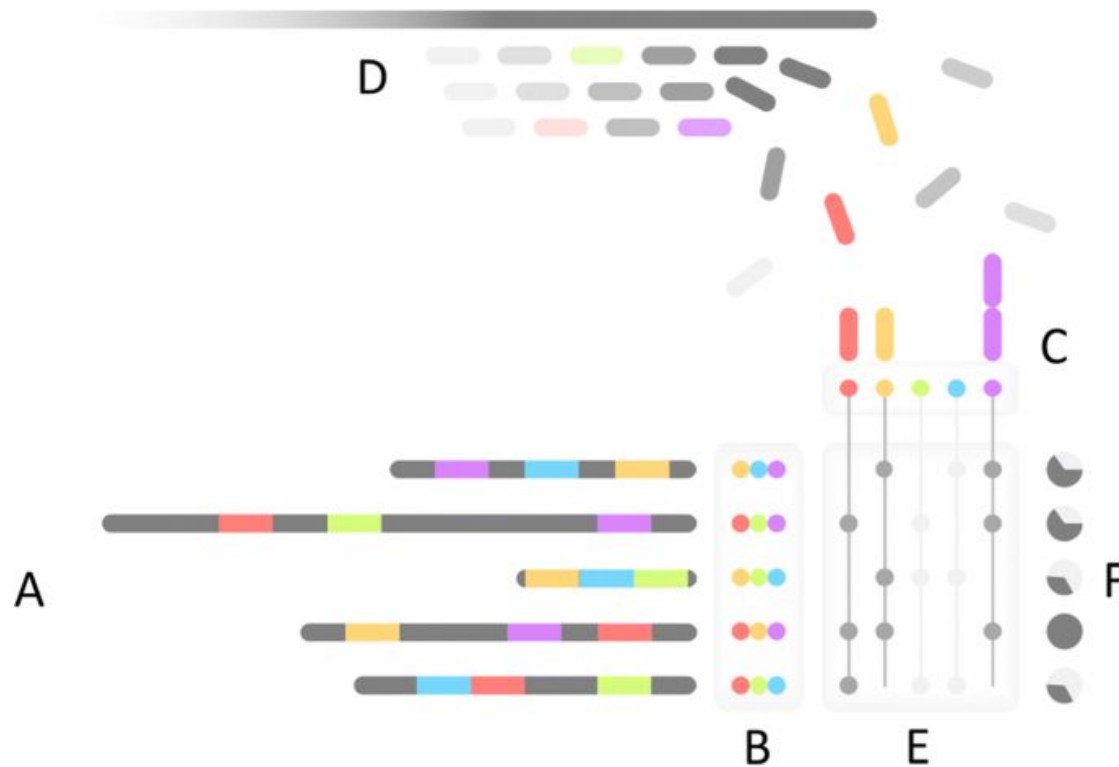


Reference based alternatives I.

- Species / genome screening using short kmers:
 - Extract short (16-31) oligonucleotide sequences from the reference genomes
 - Looking for exact matches
 - Length of the kmer can modify sensitivity and specificity
 - Covers only the fraction of the whole genome (0.5-1%)
 - Quick and low resource requirement
 - Suitable for pre-filtering, reference database creation
- Reference based gene set aligners / assemblers -> ARIBA (for AMR)

Reference based alternatives I.

- Species / genome screening using short kmers:



Reference based alternatives II.

- Extract clade / species / strain specific marker sets:
 - Align short reads to marker gene sequences
 - Derive taxon presence absence from matching marker genes
 - Derive abundance from alignment coverage
 - Build up the final abundance and taxa table
 - MetaPhlAn 4
- Protein sequence-based alignment
 - Translate read in 6 frames
 - Align to NCBI non-redundant protein database (or uniprot)
 - Diamond and MEGAN pipeline

Interpretation of results

- The complete process creates a lot of information
 - detection thresholds
- Your results are as good as your (choice of) references / markers
 - More is not always better

Thanks for your attention!